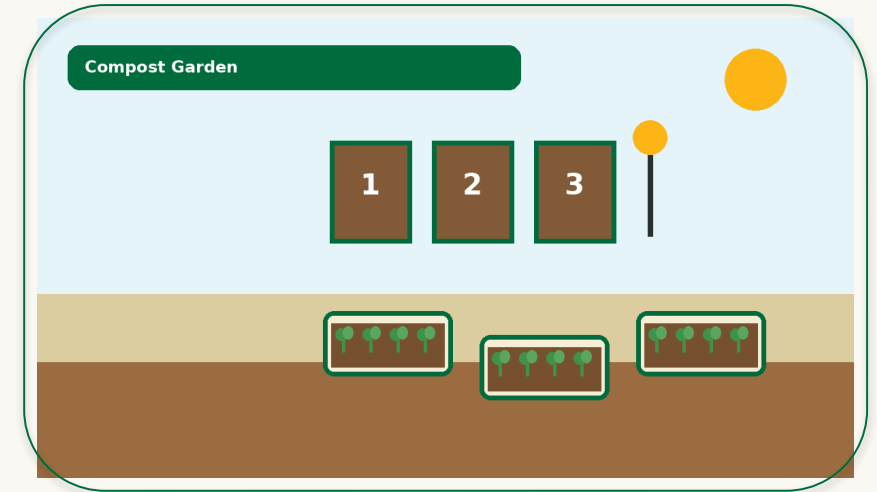


Compost Garden

Organic Waste to Soil Resource

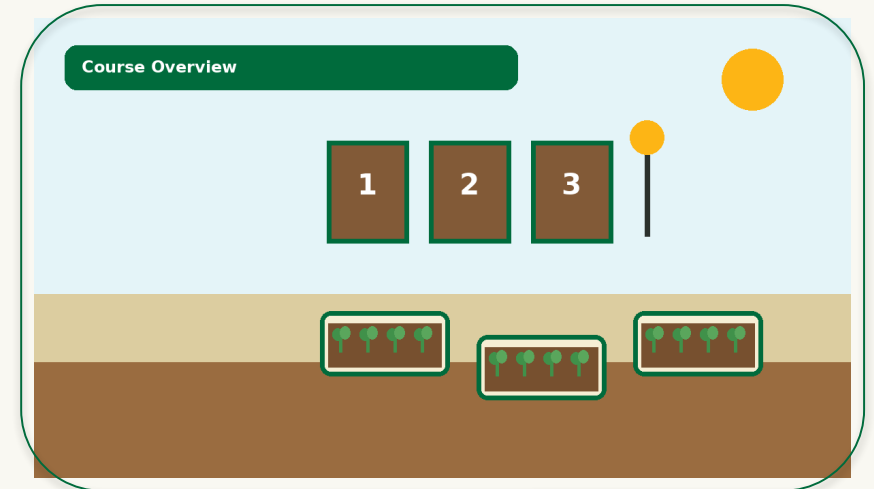
- LIFEWS AgriShine™ GrowMeal™ Primary 4-6 Teacher Training
- Audience: Primary 4-6 teachers, teaching assistants and garden facilitators
- Format: Self-paced online course + supervised garden practical
- Recommended duration: 3 hours online + 90 minutes practical
- Course emphasis: measure -> experiment -> record -> analyze -> improve



Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the garden system and its educational purpose; teach the ten technical topics at Primary 4-6 level; connect mathematics/science/agriculture to practical work; supervise measurements and garden tasks safely; maintain usable records; guide pupils from observation toward evidence-based conclusions.
- Core materials: teacher guide, crop plan, child-safe tools, watering equipment, labels, clipboards, rulers/tapes, scales or thermometers where relevant, observation/data sheets, clean containers, handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe practical lesson, and submit a short lesson/measurement or investigation plan.



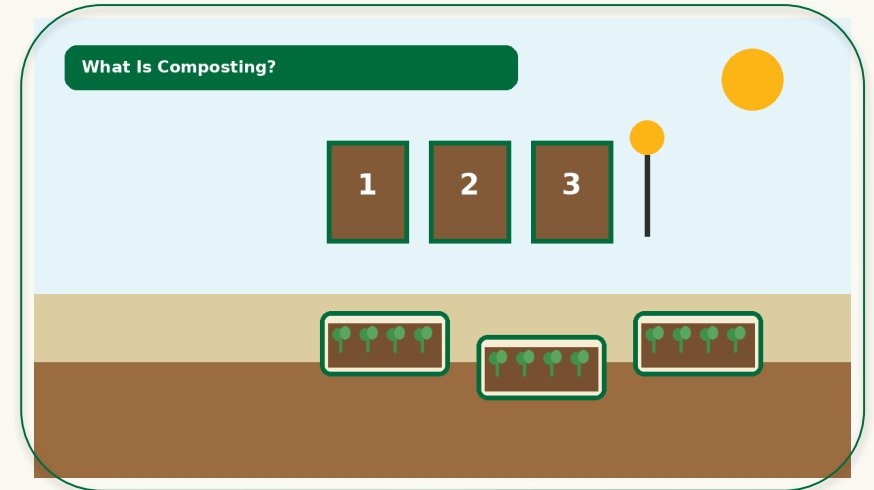
TEACHER PRACTICE

- Training sequence: EXPLAIN -> DEMONSTRATE -> MEASURE -> RECORD -> INTERPRET.

What Is Composting?

Teacher knowledge and practical application

- Define composting as managed biological decomposition of suitable organic materials into a stable soil amendment.
- Explain the roles of microorganisms, air, moisture and suitable materials at an age-appropriate level.
- Distinguish composting from simply piling waste.
- Use the school's designated compost system only.
- Connect finished compost to soil management rather than presenting it as soil itself.



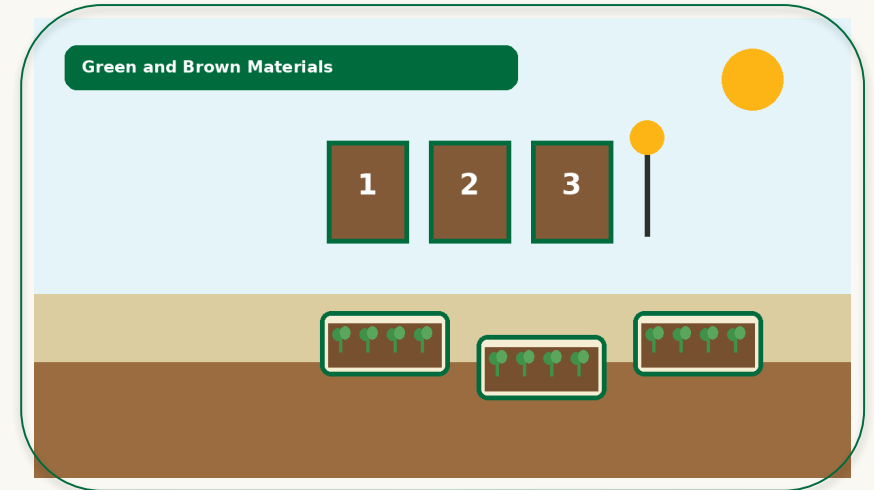
TEACHER PRACTICE

- Prepare a simple decomposition process diagram.

Green and Brown Materials

Teacher knowledge and practical application

- Introduce nitrogen-rich 'green' materials and carbon-rich 'brown' materials as practical compost categories.
- Use examples available in the school: fresh plant trimmings versus dry leaves/cardboard approved for composting.
- Explain that color does not always determine category.
- Exclude meat, dairy, oily food, hazardous materials, plastics and other prohibited items according to school protocol.
- Use a sorting activity before adding materials.



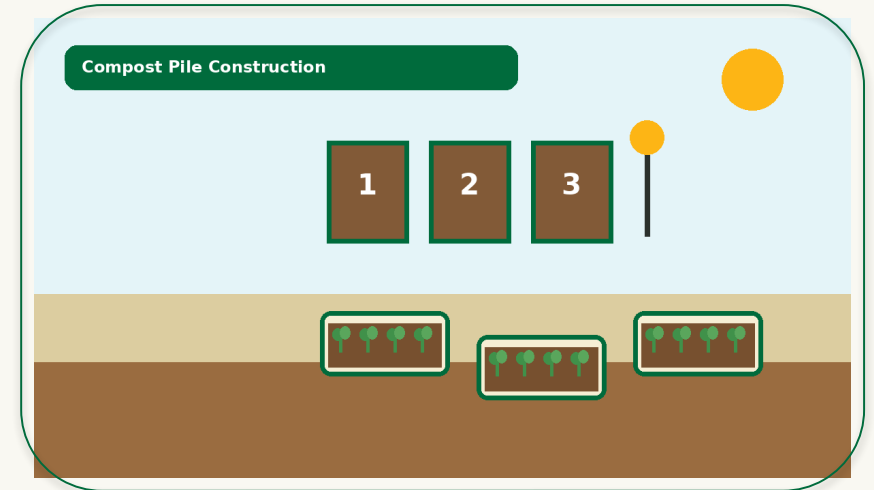
TEACHER PRACTICE

- Prepare an approved/not-approved compost sorting station.

Compost Pile Construction

Teacher knowledge and practical application

- Build compost in a designated bin or bay with adequate structure and aeration.
- Mix rather than create excessively thick single-material layers.
- Keep pile size manageable for the school system.
- Adult staff handle heavy lifting and structural repairs.
- Record the date and major material additions.



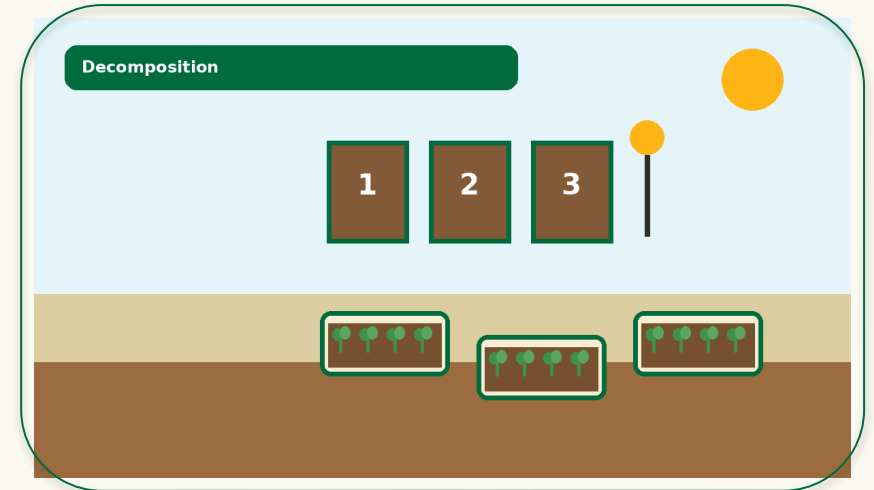
TEACHER PRACTICE

- Demonstrate safe addition and mixing of materials.

Decomposition

Teacher knowledge and practical application

- Explain that microorganisms break down organic materials over time.
- Observe changes in size, color, smell and texture without requiring pupils to handle unsafe fresh material.
- Discuss how decomposition speed depends on material, moisture, aeration and temperature.
- Avoid treating all decay as composting.
- Use photographs or mature samples for close inspection.



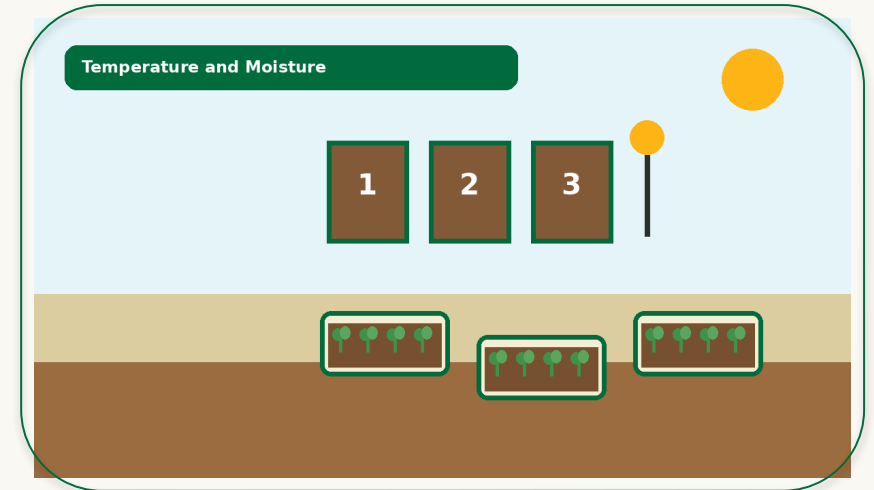
TEACHER PRACTICE

- Create a decomposition observation sequence.

Temperature and Moisture

Teacher knowledge and practical application

- Explain that active compost can become warm because of microbial activity.
- Teachers or supervised pupils may use an appropriate compost thermometer.
- Moisture should be sufficient for decomposition but not waterlogged.
- Use the 'damp sponge' comparison cautiously as a qualitative guide.
- Record temperature and moisture observations over time.



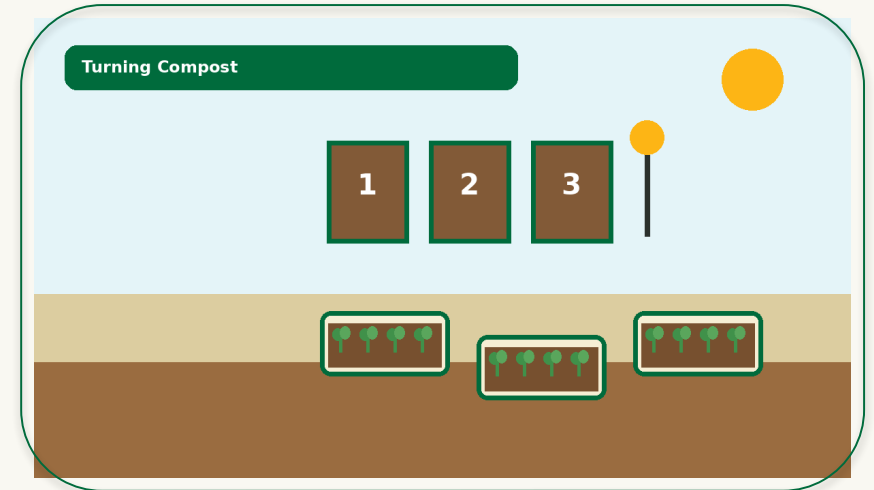
TEACHER PRACTICE

- Prepare a weekly temperature/moisture log.

Turning Compost

Teacher knowledge and practical application

- Explain turning as mixing material to redistribute moisture and introduce air.
- Turning is teacher/adult-led when forks or heavy material are involved.
- Choose frequency based on the compost system rather than a rigid rule.
- Observe temperature or odor changes after management.
- Keep pupils outside the tool-use zone.



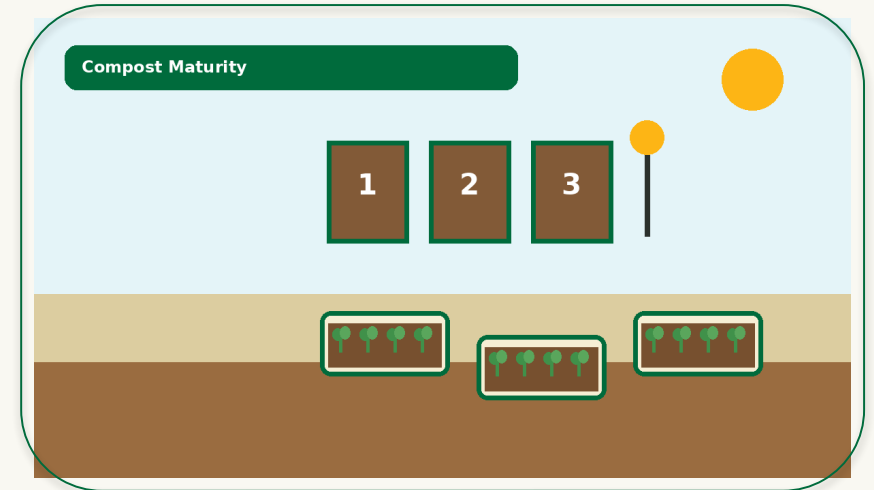
TEACHER PRACTICE

- Demonstrate safe turning-zone management.

Compost Maturity

Teacher knowledge and practical application

- Describe mature compost as dark, crumbly and earthy-smelling with original materials mostly unrecognizable.
- Explain that appearance alone is not a perfect test of maturity.
- Do not apply hot, foul-smelling or clearly unfinished material to pupil food-growing beds.
- Follow the garden program's maturity guidance.
- Store finished compost cleanly.



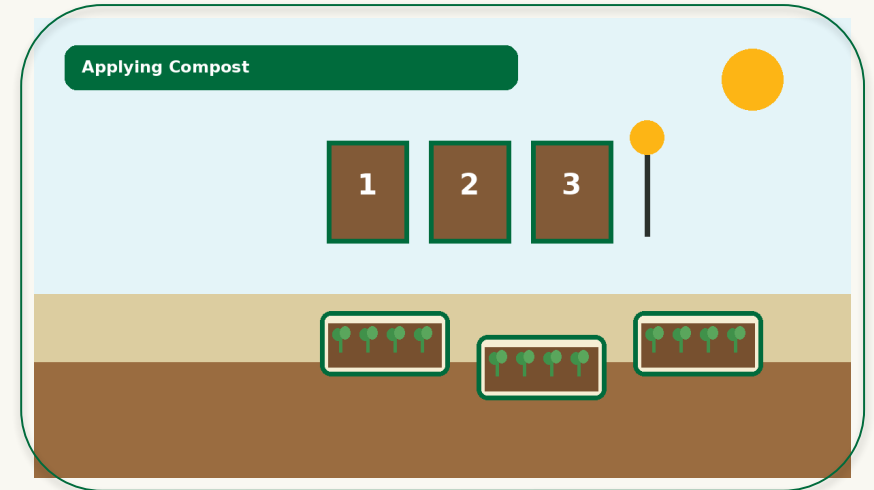
TEACHER PRACTICE

- Compare fresh inputs, active compost and mature compost using safe samples/images.

Applying Compost

Teacher knowledge and practical application

- Apply mature compost to beds or containers at rates appropriate to the garden plan.
- Mix or top-dress according to crop and system needs.
- Avoid piling compost against stems.
- Record where and when compost was applied.
- Wash hands and clean tools after use.



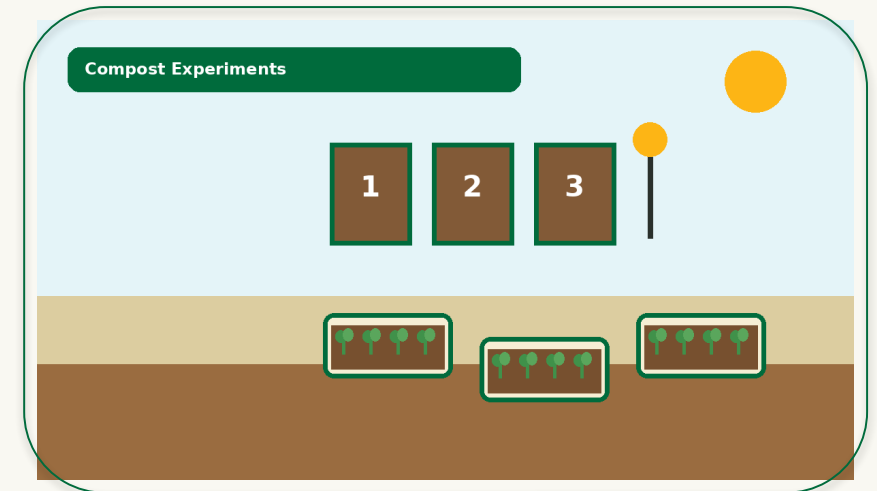
TEACHER PRACTICE

- Prepare a measured compost-application demonstration.

Compost Experiments

Teacher knowledge and practical application

- Use safe comparisons such as plant growth in approved media with different mature-compost proportions.
- Define treatments and keep other conditions similar.
- Measure growth using a consistent method.
- Do not use fresh compost as a pupil experiment treatment.
- Keep conclusions limited to the class trial.



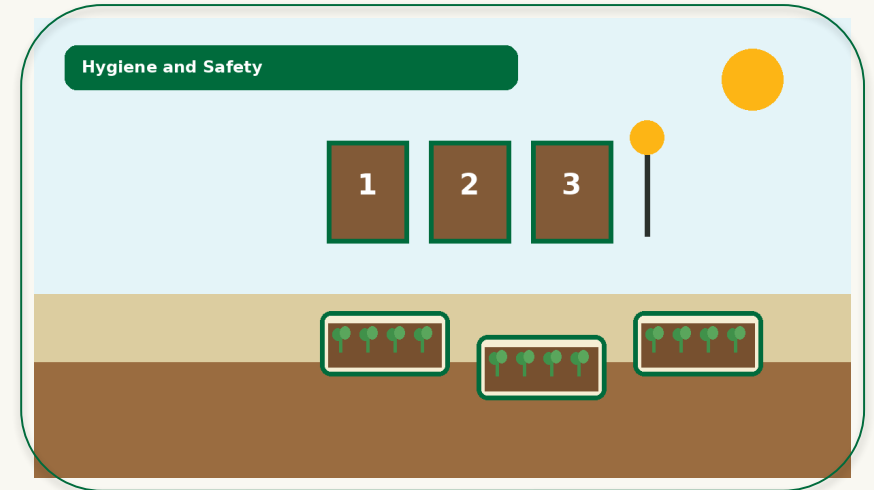
TEACHER PRACTICE

- Design a simple mature-compost comparison experiment.

Hygiene and Safety

Teacher knowledge and practical application

- Use only approved compost inputs and keep animal feces, hazardous waste and contaminated materials out of pupil systems.
- Wear appropriate gloves when required by school procedures and wash hands thoroughly after compost activities.
- Keep pupils away from sharp tools, heavy turning and hot active piles.
- Control pests and standing water through good management.
- Follow school food-safety and first-aid procedures.



TEACHER PRACTICE

- Completion task: prepare a Compost Garden safety SOP and 40-minute lesson.